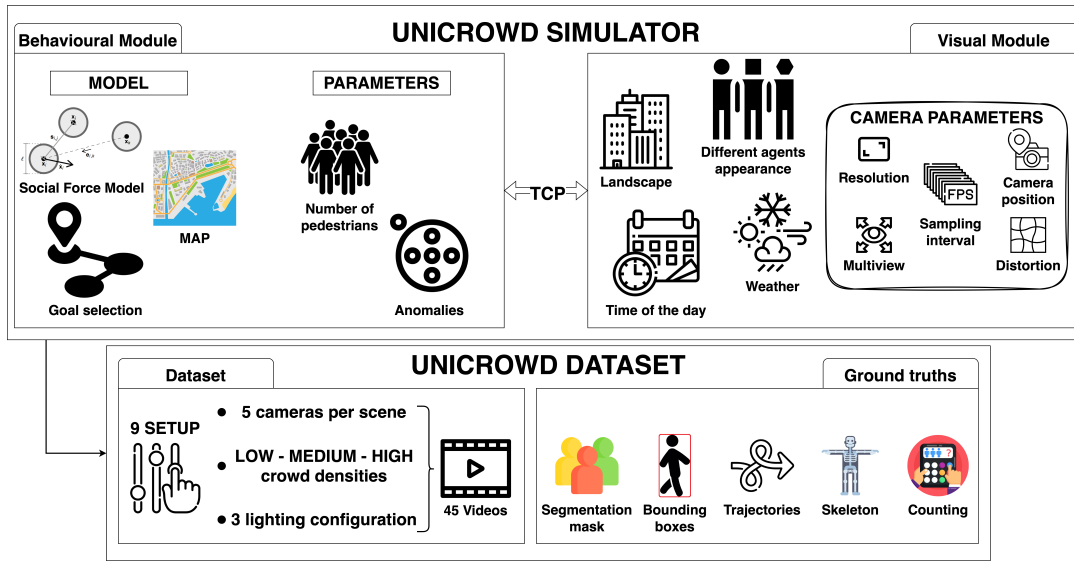


# UNICROWD SIMULATOR: VISUAL AND BEHAVIORAL FIDELITY FOR THE GENERATION OF CROWD DATASETS

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**Fig. 1:** UniCrowd is a crowd simulator consisting of 2 fully customisable modules, the former implementing the behavioral model and the latter providing the visual engine for graphical representation. Using the simulator, the UniCrowd dataset is collected, consisting of 45 video sequences in varying conditions with multiple ground truths. Our experiments demonstrate that the collected data uphold both *visual* and *behavioral fidelity*.

## ABSTRACT

We introduce UniCrowd<sup>1</sup>, a human crowd simulator for the modeling of human-related dynamics. The simulator is accompanied by a meticulously collected dataset within its synthetic environment, along with a comprehensive validation pipeline. Leveraging simulation as a powerful tool for generating annotated data, UniCrowd addresses the increasing demand for large training datasets, mimicking both the behavioral and visual aspect of crowds. Recent advancements in rendering and virtualization engines have enhanced the simulators capabilities to represent complex scenes, encompassing environmental factors such as weather conditions, surface reflectance, and human-related events like actions and behaviors. The adaptability and the non-deterministic nature of the human behavioral module of UniCrowd, coupled with its 3D rendering represents an improvement over available crowd

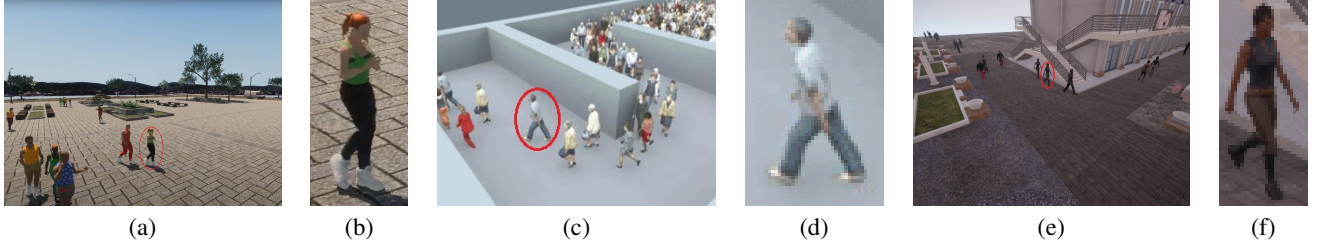
simulators. We demonstrate the suitability of our simulator and its associated dataset for various computer vision tasks. We highlight applications such as detection and segmentation, as well as specialized tasks including crowd counting, human pose estimation, trajectory analysis and prediction.

**Index Terms**— Simulation, Synthetic Data, Crowd Simulator, Pedestrian

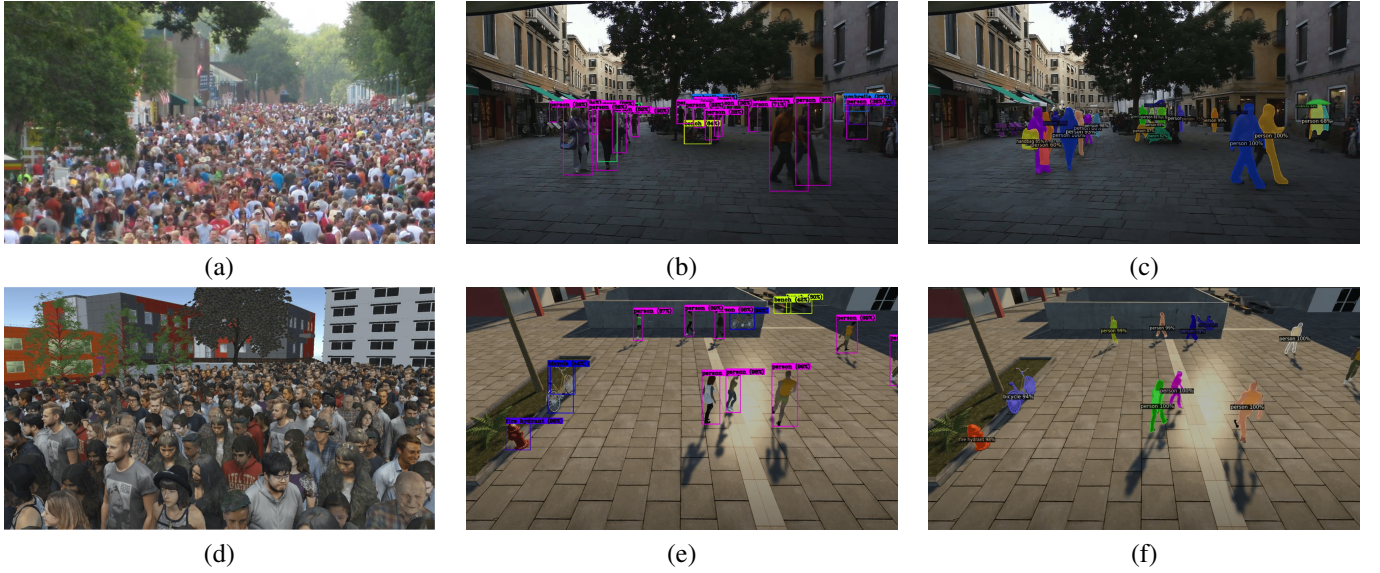
## 1. INTRODUCTION

The demand for data to train machine learning and deep learning algorithms continues to grow. As new challenges arise, the shortcomings in size, diversity and annotations of available datasets become apparent [3]. Addressing the data starvation problem necessitates a data-centric perspective. Common strategies like fine-tuning and data augmentation aim at either reconfiguring networks for different tasks or creating

<sup>1</sup>The simulator and the dataset can be accessed at [github.com/mmlab-cv/UniCrowd](https://github.com/mmlab-cv/UniCrowd)



**Fig. 2:** Visual comparison of the proposed (a, b) simulation framework against (c, d) Agoraset [1], and (e, f) LCCrowdV [2] datasets. Agoraset lacks realism; LCCrowdV has low resolution, limited ground truth data for different tasks, and low-quality rendering (e.g. light simulation, shadows).



**Fig. 3:** Crowd analysis techniques applied to real crowd data (a, b, c) and synthetic data (d, e, f) simulated with the UniCrowd simulator. The UniCrowd simulator can generate video sequences and the ground truth data suitable for a number of tasks, as crowd counting (d), people detection (e) and people segmentation (f). A multi-purpose simulation framework must be able to reproduce both behavior and appearance of crowds, suitable for multiple tasks.

new instances through transformations applied to available samples. Both strategies try to avoid annotating new samples, which would require intensive human labour. In image processing and computer vision, the annotation cost is particularly high, since images and videos are annotated one by one, often requiring the manual selection of the relevant regions of interest. This may lead to the so-called *curse of dataset annotation* [4]: when datasets tend to be smaller, the more detailed the annotation is. Viceversa, when the dataset size is huge, the annotations are less accurate or have inconsistent labels [5].

Among the tools to address the lack of annotated data, researchers have explored the chance of replacing (or complementing) the collected samples by using simulation frameworks [1, 6, 7]. Simulators provide accurate ground truth data, reducing the time required for annotation and enabling the collection of bigger and exhaustive datasets. However,

relying on simulators might introduce additional issues, as the generated (synthetic) data must closely resemble their real counterpart.

Simulators should then ensure that the generated data comply with *visual fidelity*, and, when humans or moving agents are involved, *behavioral fidelity*.

*Visual Fidelity* assesses how close the synthetic scene visually resembles a real-world one, as captured by a real camera. A given algorithm should perform comparably when trained with either real or synthetic data generated with a simulator. Thus, one of the goals of a simulator is to provide data that allow a (possibly) seamless transfer between the synthetic domain and the real one.

*Behavioral Fidelity* refers to the ability of the simulator to model the dynamics of the target domain in a coherent and realistic way. This requires the observation of real-world behavioral dynamics, which must be consequently modeled in

a synthetic scene. This is particularly relevant when dealing with videos, to guarantee temporal consistency across successive frames.

With this respect, crowd analysis provides a rich and diversified use case scenario, in which simulators can play a relevant role: the scene should replicate the appearance of a crowd, which consists of multiple subjects exhibiting different appearance and behaviors. These elements imply fulfilling the requirements of both *visual fidelity* and *behavioral fidelity*, simulating and modeling the visual rendering, as well as the ongoing social interactions.

UniCrowd, along with its associated dataset, is a comprehensive simulator framework, designed to address common computer vision tasks like detection and segmentation, and more behavior-centric challenges such as crowd counting, human pose estimation, trajectory analysis, and prediction, as shown in Fig. 1.

Unlike many existing crowd simulators [1, 2, 8], which often prioritize behavioral simulation at the expense of visual rendering quality, UniCrowd stands out by excelling in both *visual fidelity* and *behavioral fidelity*. This superiority is evident in Fig. 2, showcasing UniCrowd's ability to surpass the visual fidelity limitations observed in other simulators.

Moreover, UniCrowd allows users to generate data with full customization control, including the number of cameras, their positions and resolutions, as well as the layout and features of the simulated environment. This versatility is reflected in our collected UniCrowd dataset, comprising 45 videos featuring diverse crowd densities, multiple cameras, and varied environmental configurations. Additionally, the dataset incorporates multiple ground truths for a variety of tasks, surpassing the richness of features found in existing synthetic crowd datasets [6, 9, 10] as shown in Fig. 3.

## 2. RELATED WORK

The use of synthetic datasets in computer vision is not new per se [7], and the recent advancements in the video-games industry have enabled the development of improved and highly desirable graphical representations.

Through simulation, it is possible to create large datasets consisting of images, videos, metadata, comprising of accurate and automatically-generated ground truth, in the form of bounding boxes, per-pixel depth, optical flow, semantic classes and instance segmentation, exploiting video games-like paradigms [7].

The use of computer-generated data has motivated researchers in investigating the effectiveness of the synthetic data (see for example [10]), and how much strategic they are in solving real-world problems. For this reason procedural generation of synthetic videos [15] as well as the integration of 3D engines [16] have been widely exploited, and the available approaches aim at creating tools capable on the one hand to generate realistic scenes, and on the other hand produce the

related ground truth. More recent approaches have employed modern video-game rendering to obtain the best possible appearance [6, 11]. In the crowd analysis domain, Agoraset [1] and LCrowdV [2] have introduced visual simulations to create synthetic datasets. LCrowdV [2] focuses on validating the *visual fidelity* on people detection. Agoraset [1] focuses on validating the *behavioral fidelity*, providing validation for the generated trajectories. In our work, we tackle both aspects, aiming at consolidating both the visual appearance features and the behavioral modules of crowd simulators. An overview of synthetic simulators and datasets in the domain of crowd analysis is shown in Tab. 1.

## 3. UNICROWD SIMULATOR

UniCrowd is conceived to meet the requirements of *behavioral fidelity* and *visual fidelity*. It consists of two main blocks, a behavioral and a visual module, seamlessly communicating through a TCP socket. This modular structure not only ensures the robustness of each component but also facilitates their independent replacement and updates. This adaptability is designed to accommodate future developments and integrations, providing a dynamic and scalable platform.

**Behavioral module.** On the behavioral side, our solution is based on the Social Forces Model (SFM) [17]. The SFM is an agent-based model, where each agent computes its next position given its current velocity, and repulsive forces with respect to obstacles and other pedestrians. SFM has proven effective in modeling the micro behaviors of the crowd, while retaining an overall accurate overview of events at a macroscopic level. Compared to other solutions [18, 19], which can resemble the crowd behavior in well defined contexts, the SFM can generalise to a multitude of scenarios, thus ensuring enhanced flexibility, especially in dealing with environmental changes. Moreover, it can be refined and enriched with more sophisticated behaviors like group formations [20].

The behavioral section of the simulator is decoupled from the graphical part: given the agent's position at time step  $t$ , and considering its final destination and other personal parameters, it produces the next position at time step  $t + 1$ . The position is then fed to the graphical rendering block using the TCP socket.

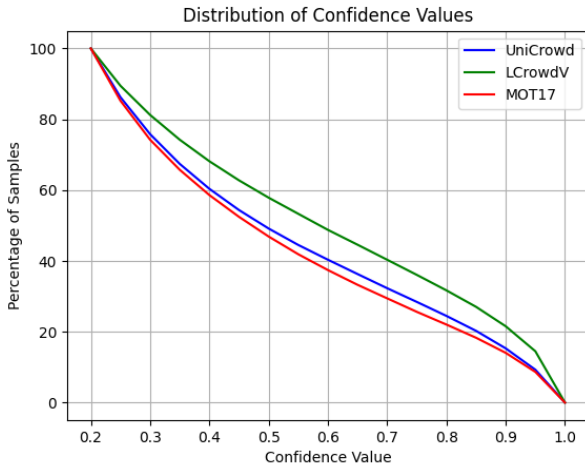
Unlike other simulators that respond to pre-computed scripts [8, 2, 1], having the two macro-blocks separated opens to the possibility of controlling the simulation in a dynamic fashion, generating events and anomalies.

The main parameters controlled by the module are the number of pedestrian in the simulation, their spawning and target points, and the repulsive forces, which model the interpersonal distances, and their preferred velocity.

**Visual module.** The graphical rendering is handled by Unity3D, using available assets for the generation of both the crowd and the environment. As shown in Fig. 2, available simulators like Agoraset [1] and LCrowdV [2] also provide

|           | Methods       | Ground Truth |              |            |            |          | Behavioral Fidelity | Visual Fidelity |
|-----------|---------------|--------------|--------------|------------|------------|----------|---------------------|-----------------|
|           |               | Segmentation | Bounding Box | Trajectory | Human Pose | Counting |                     |                 |
| Dataset   | JTA [6]       | ✗            | ✗            | ✗          | ✓          | ✗        | ✗                   | ✓               |
|           | MotSynth [11] | ✓            | ✗            | ✗          | ✓          | ✗        | ✗                   | ✓               |
|           | Viper [12]    | ✓            | ✗            | ✗          | ✓          | ✗        | ✗                   | ✓               |
|           | GTA [13]      | ✓            | ✗            | ✗          | ✗          | ✗        | ✗                   | ✓               |
|           | PFD [7]       | ✓            | ✗            | ✗          | ✗          | ✗        | ✗                   | ✓               |
|           | Synthia [9]   | ✓            | ✗            | ✗          | ✗          | ✗        | ✗                   | ✓               |
| Simulator | Menge [8]     | ✓            | ✗            | ✗          | ✓          | ✗        | ✓                   | ✗               |
|           | AgoraSet [1]  | ✗            | ✓            | ✓          | ✗          | ✓        | ✓                   | ✓               |
|           | LCrowdV [2]   | ✗            | ✓            | ✓          | ✗          | ✓        | ✓                   | ✓               |
|           | <b>Ours</b>   | ✓            | ✓            | ✓          | ✓          | ✓        | ✓                   | ✓               |

**Table 1:** Synthetic datasets comparisons; a complete overview can be found at [14]. The first six methods provide synthetic datasets for deep learning applications, which are generated without validating the behavioral fidelity of the involved agents. The last three methods also provide full simulation engines, which can be used by researchers to generate new data. They also validate a behavioral model. As can be seen, our proposal provides the most complete setup.



**Fig. 4:** To evaluate the visual fidelity of the proposed simulator, we run an off-the-shelf version of YOLO on our dataset, LCrowdV, and, for comparison, the video sequences from the MOT17 challenge. In this graph, we show the percentages of bounding boxes with a confidence above the threshold on each dataset. Comparing against real videos from the MOT challenge (red), it can be seen that UniCrowd performs similarly to the real videos, while LCrowdV (green) appears to be less accurate in replicating realistic contents.

graphical rendering for their scenarios. However, Agoraset has a low-quality rendering, especially as far as the environmental features are concerned. LCrowdV, instead, suffers of small image size and bad illumination handling. A qualitative comparison is shown in Fig. 2, showing how UniCrowd provides an improvement in camera resolution and image size. A quantitative comparison is shown in Fig. 4, where it can be seen how the visual appearance of UniCrowd leads to a confidence distribution of human detection, which closely resembles the distribution on real videos with similar content [21].

Moreover, our simulator is rich in features to diversify the simulation, and to provide diverse scenarios, in particular with respect to:

- global dynamic illumination. It is possible to change the illumination depending on the time of the day (e.g. light fading and changing colors at sunset);
- weather conditions. Multiple different weather condition can be simulated, like rain and snow;
- camera parameters with real world distortion filters. Our rendering cameras are modeled to simulate the recording using real lenses (e.g. adding flares and barrel distortion) and real sensors (e.g. granular noise). Lenses, resolution, and focal length can be customised and can vary for each camera, simulating a real-world system where not all cameras are homogeneous and different cameras can be placed and used differently according to their characteristics;
- position and number of cameras, each with its relative ground truth data;

- diverse agents appearance, in terms of gender, clothes, and body shape.

#### 4. UNICROWD DATASET

| Low-level          | Mid-level       | Global             |
|--------------------|-----------------|--------------------|
| Bounding boxes     | Trajectories    | Weather and time   |
| Joints' position   | Anomalies       | Crowd densities    |
| Segmentation masks | People counting | Spawn/target areas |

**Table 2:** Provided ground truth annotations for low and mid-level tasks, and global configuration parameters.

From the simulation engine, we have collected a comprehensive dataset, showcasing its ability to provide ground truth data for various tasks encompassing both appearance and behavioral features. This comprehensive dataset caters to a variety of tasks, with some heavily relying on the behavioral aspects of pedestrians in the simulator (e.g., trajectories and anomalies), while others concentrate on visual appearance (e.g., segmentation masks and crowd counting). The ground truth annotations divided in low and mid-level tasks, along with global configuration parameters, are detailed in Table 2.

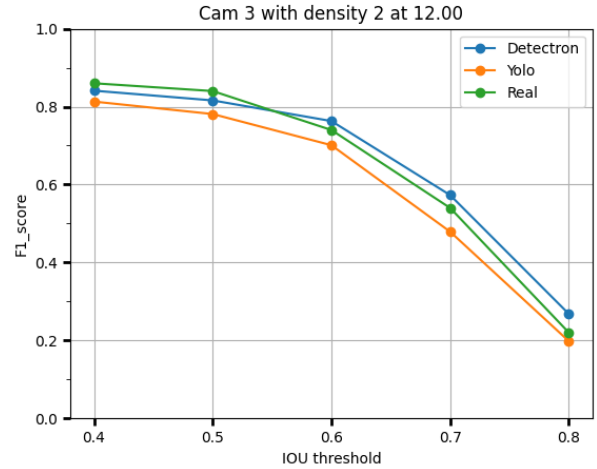
The dataset comprises scenes recorded by five different cameras, each capturing the same scene under nine distinct environmental conditions. The variations include variable crowd density and overall number of people (40, 100, and 150 people per scene, respectively) and diverse lighting conditions (recorded at 7 am, 12 pm, and 6:30 pm). This results in a total of 45 Full HD videos at 25 Fps with an average length of 2 minutes for a total of 130k annotated frames, creating a rich and diverse set of scenarios. We provide a around 2000 unique trajectories, with each pedestrian having its own bounding box and segmentation mask for each camera.

We also provide accurate positioning data: 3D maps and positions, 2D positions on the image plane of each camera, as well as annotated joints, trajectories, segmentation mask and bounding box. Moreover tasks such as re-identification and multiple camera tracking can be tested, since we provide consistent pedestrian ID across the whole scene.

#### 5. EXPERIMENTS

The provided ground truth data can be used for a variety of tasks/applications, as shown in Fig. 3. For the validation of the proposed pipeline, we focus on people detection and segmentation as they represent two key use-cases, on top of which, multiple scenarios can be envisaged.

The most common methods in literature target either robustness and accuracy of the detection and segmentation [22] or they explore the trade-off between accuracy and computational burden [23].



**Fig. 5:** F1-scores obtained varying the IOU threshold on the Cam3 at 12:00, with a medium density crowd with YOLO (orange line) and Detectron (blue line). YOLO [24] applied to real data [25] with similar viewpoint to the synthetic one (green line) performs similarly to simulated data (orange line).

**People detection.** In the detection domain [21], we define for the evaluation a varying threshold of the Intersection over Union (IOU) at which we compute the  $F1$  score. The  $F1$  is used as the metric to evaluate the performances of the off-the-shelf versions of YOLO (detection) [23] and Detectron [22] on the synthetic data.

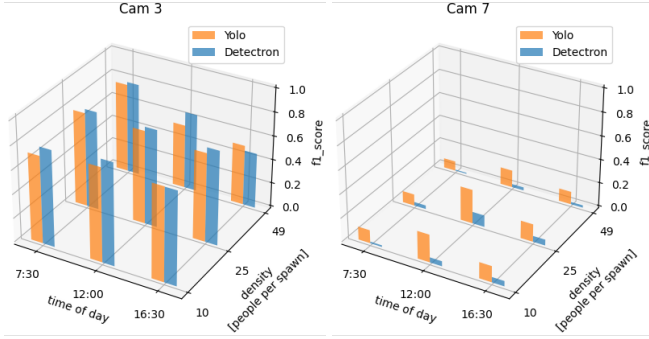
In Fig. 5, we report results for both YOLO and Detectron with a varying threshold between [0.4, 0.5, 0.6, 0.7, 0.8].

In Fig. 6, we provide an overview of how the performances vary for two selected cameras, at different crowd densities and time of the day. Both algorithms obtain good results when dealing with the side-viewpoint (Fig. 6(a)), but struggle when dealing with the top-viewpoint (Fig. 6(b)).

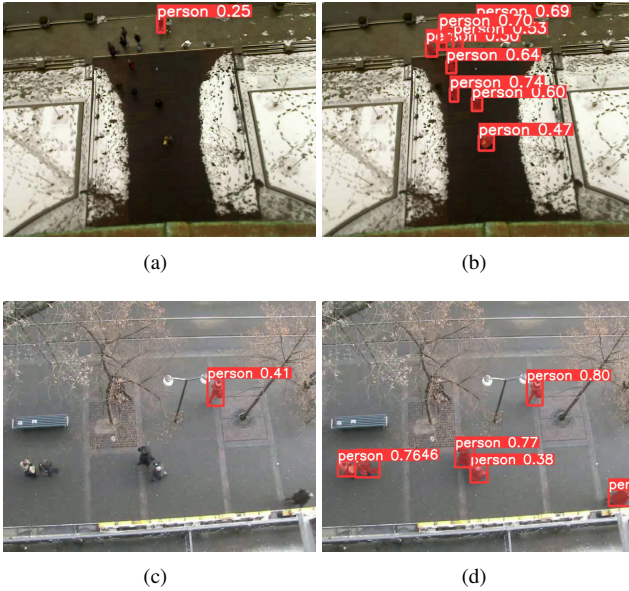
The top view-point presents the most challenging scenario for state-of-the-art algorithms, as shown in Fig. 7(a,c) and in Fig. 6, because of the scarcity of training data. Simulated data can then be of help with this respect, providing sufficient amount of data from any arbitrary viewpoint. In Fig. 7(b,d), our experiments show how performing a fine-tuning with extra synthetic data allows for a huge jump in detection performance when moving into the real world scenario.

**People segmentation.** In the segmentation domain, we evaluate the performance of Detectron [22] using the standard COCO metrics [26], which evaluates AP (Average Precision) for object detection and instance segmentation. For COCO, it is implied that AP is an average over all the categories, which is better referred as mean average precision (mAP).

The obtained results are reported in Tab. 3, showing comparable performances of the off-the-shelf model trained on real data and tested on both real and synthetic data.



**Fig. 6:** F1-scores obtained applying off-the-shelf version of Detectron and YOLO on two different views ((a) Cam3, (b) Cam7) from synthetic data, varying the people density and the time of the day. Cam3 corresponds to a frontal viewpoint; Cam7 corresponds to a quasi top-view and shows how off-the-shelf algorithms struggle to obtain good results due to the lack of training data from such viewpoint.



**Fig. 7:** Challenging viewpoints detection can benefit from synthetic data augmentation. When applied on top-view images ((a) and (c)), the off-the-shelf version of YOLO struggles to identify people from the top view. Performing data augmentation with synthetic data provides much better results ((b) and (d)).

|           | AP     | APs    | APm    | API    |
|-----------|--------|--------|--------|--------|
| Synthetic | 58.808 | 37.761 | 62.104 | 73.981 |
| Real      | 58.670 | 23.011 | 53.235 | 66.209 |

**Table 3:** Comparison of segmentation results obtained testing the off-the-shelf version of the Detectron algorithm on both real and synthetic test dataset.

(Keys: AP is the Average Precision, AP50 is the average Precision using an IoU threshold of 0.5, AP75 is the Average Precision using a IoU threshold of 0.75 , APs is the Average Precision on small objects , APm is the Average Precision on medium objects, API is the Average Precision on large objects)

Both detection and segmentation algorithms perform similarly on the synthetic video and in the real world, as shown in Fig. 3.

## 6. CONCLUSIONS

In this work, we have presented UniCrowd, a ready-to-use simulation tool and a rich synthetic dataset comprising data of pedestrians in a variety of environmental conditions. The simulation engine and the data are suitable for multiple applications that deal with crowd analysis from a computer vision perspective. *Visual fidelity* and *behavioral fidelity* have been highlighted as a new paradigm that can be used to evaluate the effectiveness of synthetic data. We have chosen the use case of crowd analysis to display the potential of our framework, since it provides a challenging and complete example for both the behavioral and visual models, demonstrating the applicability of such paradigm to multiple use cases, such as detection, tracking and segmentation.

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